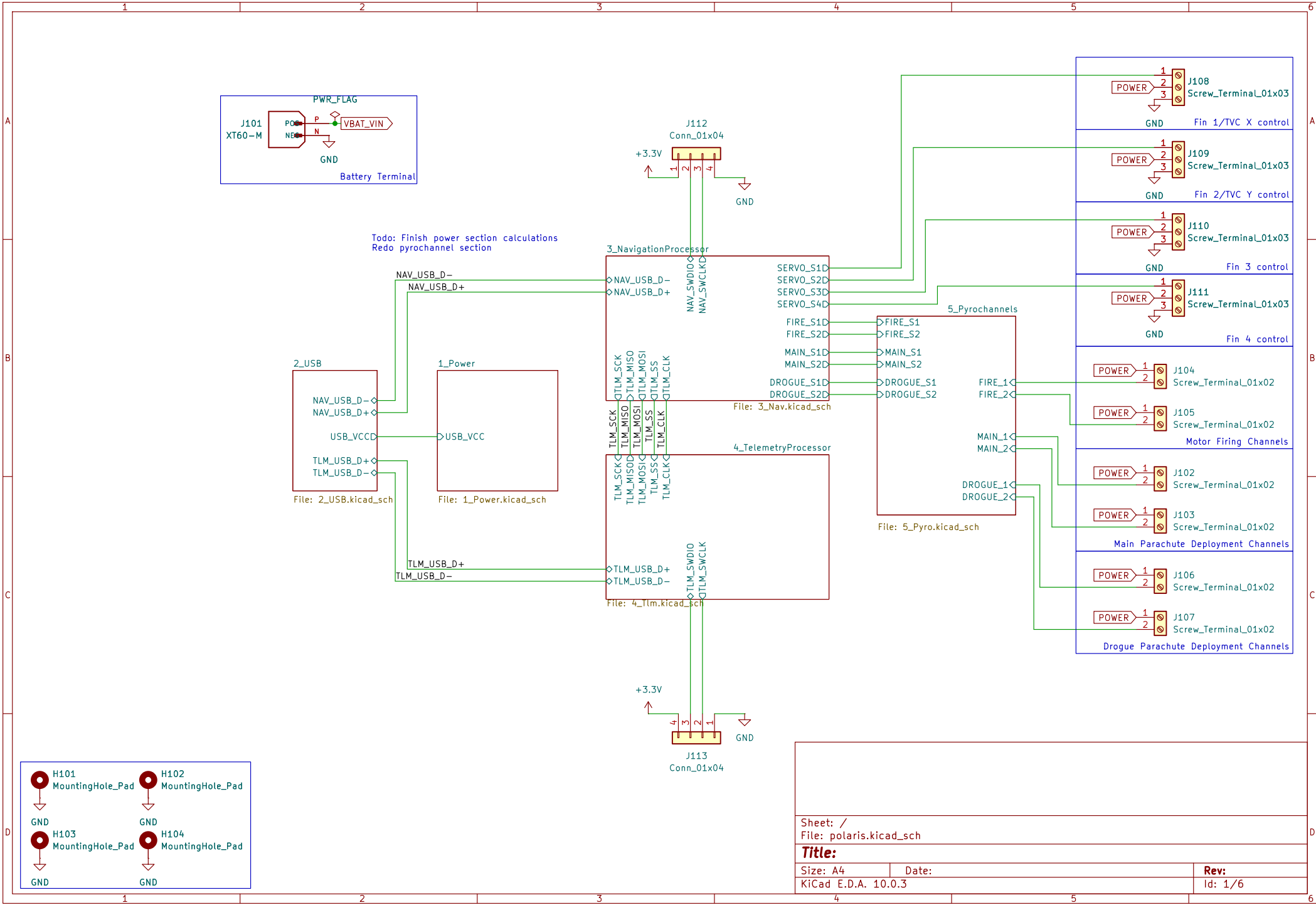
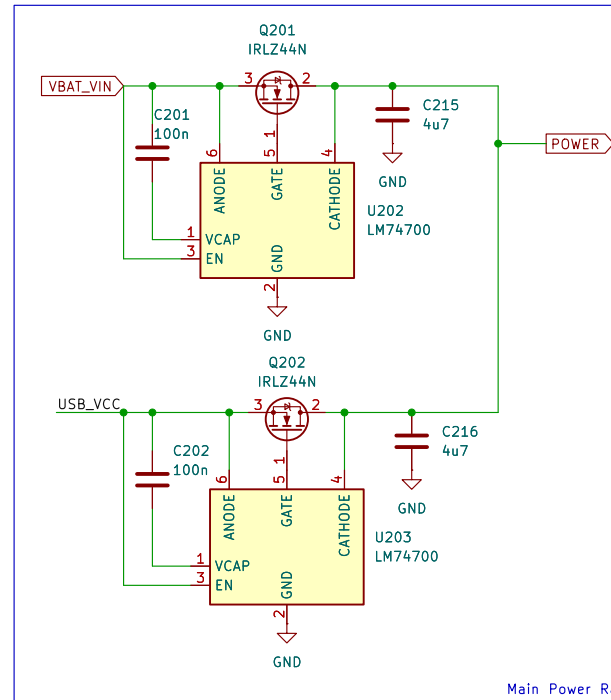
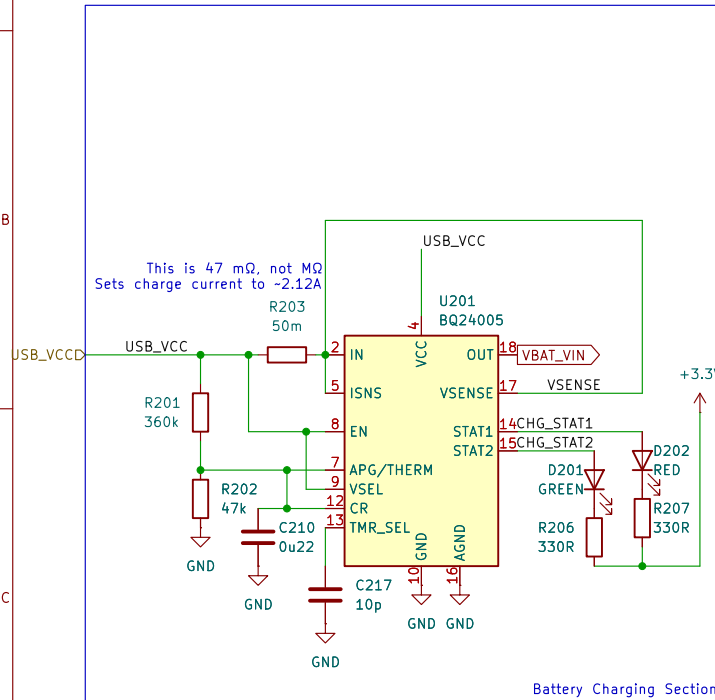




[1] Power



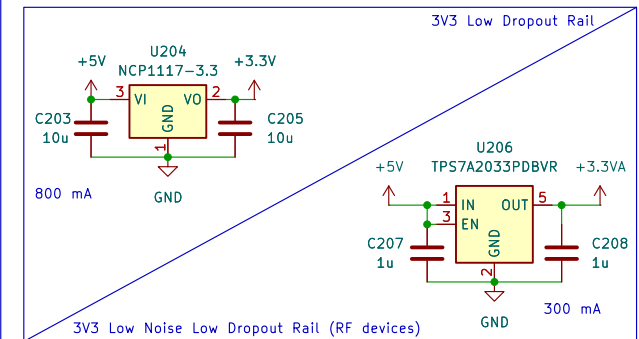
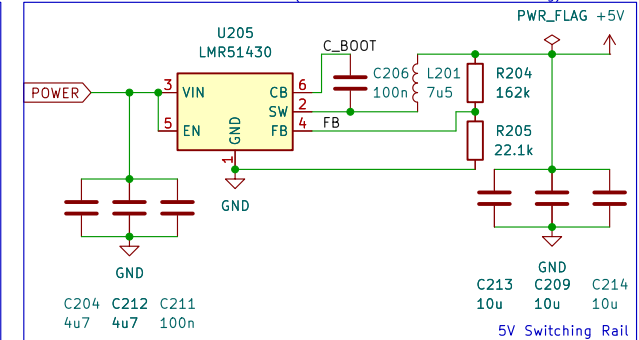
Output inductor calculations:
 $L_{min} = (V_{in_max} - V_{out}) * V_{out} / I_{out} * K * V_{in_max} * f$
 $V_{in_max} = 9V, V_{out} = 5V, I_{out} = 2A, K = 0.3, f = 500kHz$
 $L_{min} = -7.4 \mu H = 7.5 \mu H$ standard value

Output capacitor calculations:
 $\Delta V_{out_esr} = K * I_{out} * ESR = 30 \text{ mV max}$
 $\Rightarrow ESR = 0.05 \Omega \text{ max}$

$\Delta V_{out_c} = K * I_{out} / 8 * f * C_{out} = 30 \text{ mV max}$
 $\Rightarrow C_{out} = 5 \mu F$

$C_{out} > 4 * (I_{oh} - I_{ol}) / f * \Delta V_{out_shoot}$
 Using $I_{oh} = 1.5A, I_{ol} = 0.5A, \Delta V_{out_shoot} = 250 \text{ mV}$:
 $C_{out} > 32 \mu F$

Using $C_{out} > 32 \mu F$ at 50 mΩ ESR:
 Choose $C_{out} = 3 * 10 \mu F$ at 25V (CL31B106KAHNNWE from Samsung)



Sheet: /1_Power/
 File: 1_Power.kicad_sch

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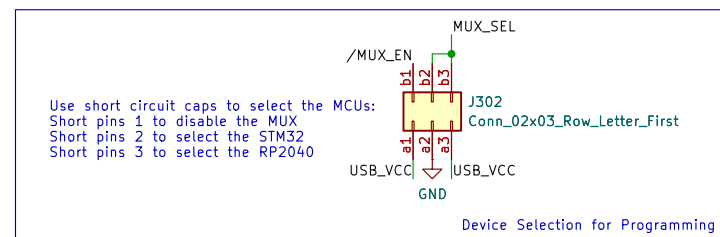
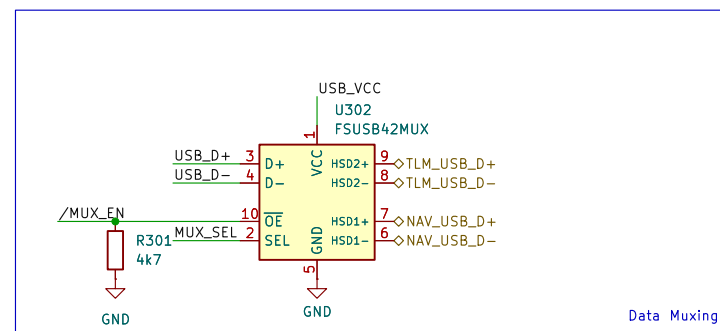
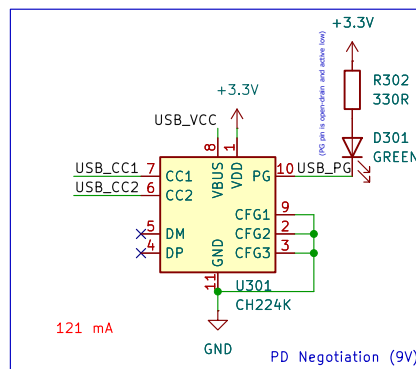
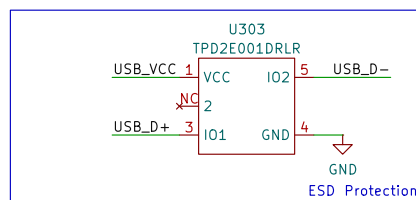
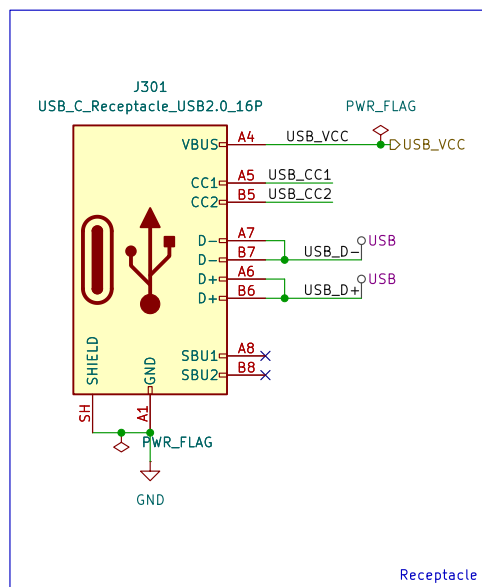
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[2] USB



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File: 2_USB.kicad_sch

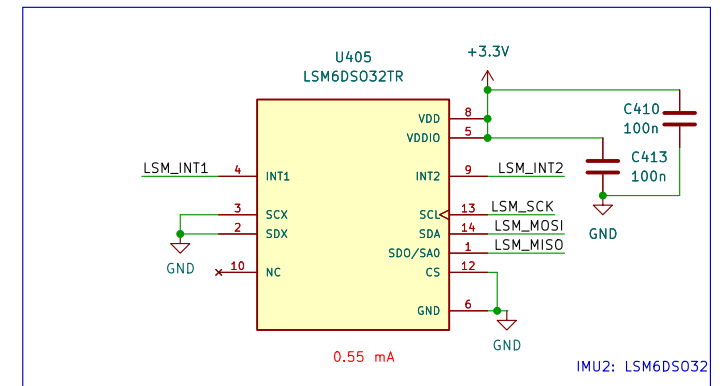
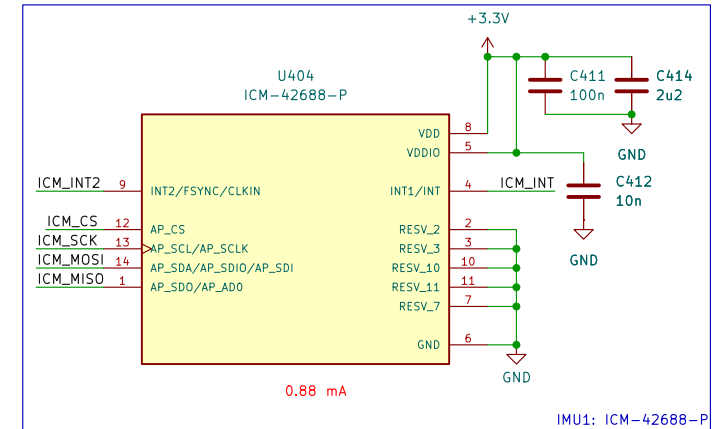
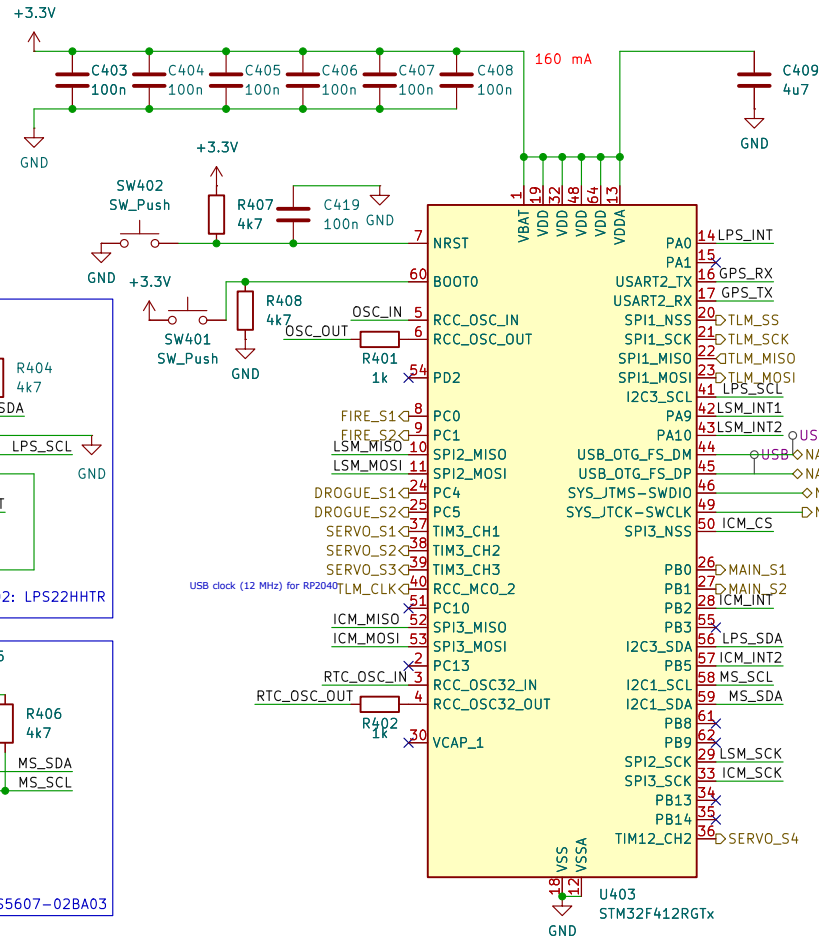
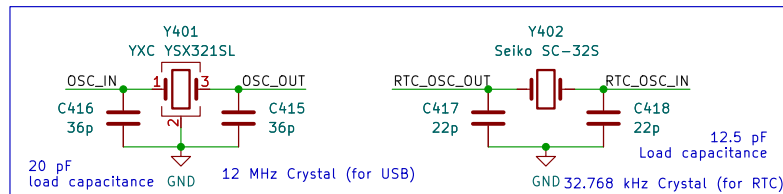
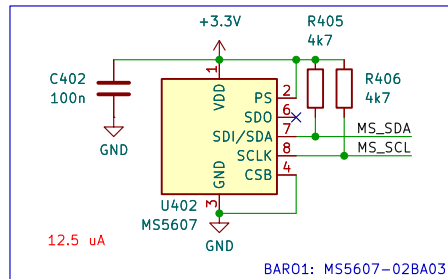
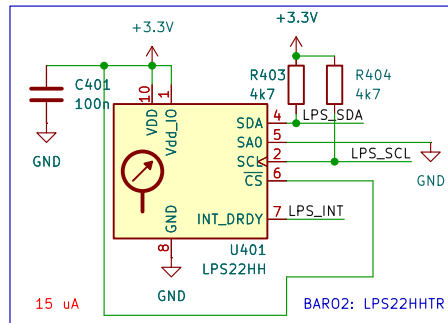
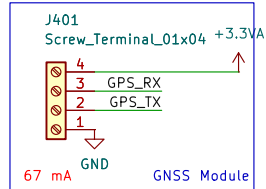
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[3] NAV



SPI1: RP2040
SPI2: LSM6DS032
SPI3: ICM-42688-P
I2C1: M55607
I2C3: LPS22HH
USART2: GPS NEO-M8N

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File: 3_Nav.kicad_sch

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[4] TLM

Hold down this button and plug the board to a computer with RPi as USB device to program it

25 mA

100 mA

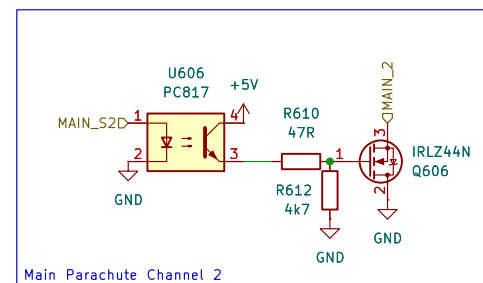
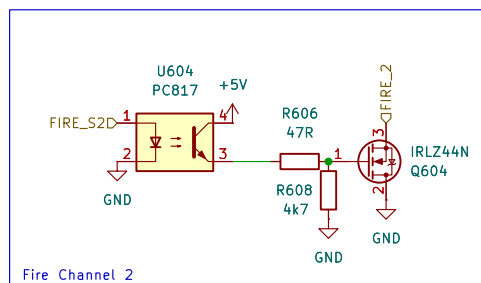
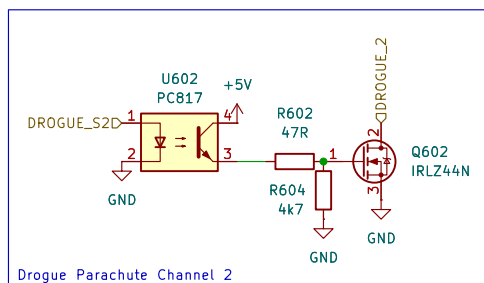
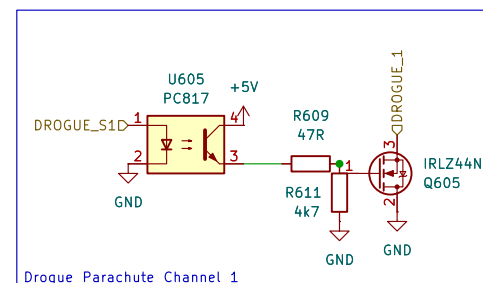
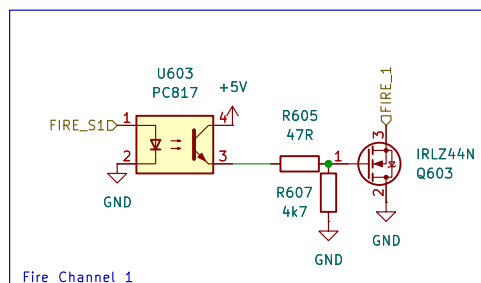
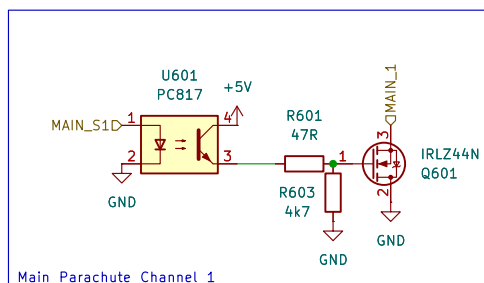
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[5] PYRO



20 mA total

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